

MATB34 - Written Assignment

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1 Laplace transforms

(1)

$$\mathcal{L}\{\cos^2(\omega t)\} = \mathcal{L}\left\{\frac{1}{2}(1 + \cos(2\omega t))\right\} = \frac{1}{2}(\mathcal{L}\{1\} + \mathcal{L}\{\cos(2\omega t)\}) = \frac{1}{2}\left(\frac{1}{s} + \frac{s}{s^2 + 4\omega^2}\right).$$

(2)

$$\mathcal{L}\{t^2 - 1\} = \mathcal{L}\{t^2\} - \mathcal{L}\{1\} = \frac{2}{s^3} - \frac{1}{s}.$$

(3)

$$\mathcal{L}\{e^{-t} \cos^2 t\} = \mathcal{L}\left\{e^{-t} \cdot \frac{1}{2}(1 + \cos(2t))\right\} = \frac{1}{2}\mathcal{L}\{e^{-t}\} + \frac{1}{2}\mathcal{L}\{e^{-t} \cos(2t)\} = \frac{1}{2(s+1)} + \frac{s+1}{2((s+1)^2 + 4)}.$$

2 Inverse Laplace transforms

(1)

$$\begin{aligned}\mathcal{L}^{-1}\left\{\frac{1}{(s-a)(s-b)}\right\} &= \mathcal{L}^{-1}\left\{\frac{1}{a-b}\left(\frac{1}{s-a} - \frac{1}{s-b}\right)\right\} \\ &= \frac{1}{a-b}\mathcal{L}^{-1}\left\{\frac{1}{s-a} - \frac{1}{s-b}\right\} \\ &= \frac{1}{a-b}\left(\mathcal{L}^{-1}\left\{\frac{1}{s-a}\right\} - \mathcal{L}^{-1}\left\{\frac{1}{s-b}\right\}\right) \\ &= \frac{1}{a-b}(e^{at} - e^{bt}) \\ &= \frac{e^{at} - e^{bt}}{a-b} \quad (a \neq b).\end{aligned}$$

$$\text{If } a = b \text{ then } \mathcal{L}^{-1}\left\{\frac{1}{(s-a)^2}\right\} = te^{at}$$

(2)

$$\begin{aligned}\mathcal{L}^{-1}\left\{\frac{s^2}{1+s^2}\right\} &= \mathcal{L}^{-1}\left\{\frac{s^2+1-1}{s^2+1}\right\} \\ &= \mathcal{L}^{-1}\left\{1 - \frac{1}{s^2+1}\right\} \\ &= \mathcal{L}^{-1}\{1\} - \mathcal{L}^{-1}\left\{\frac{1}{s^2+1}\right\} \\ &= \delta(t) - \sin t.\end{aligned}$$

3 Ordinary differential equations

(1)

$$\begin{aligned}
 y'' - y' + 2y &= 2e^{3t}, \quad y(0) = 0, \quad y'(0) = 0. \\
 (s^2Y - sy(0) - y'(0)) - (sY - y(0)) + 2Y &= \frac{2}{s-3} \\
 (s^2 - s + 2)Y &= \frac{2}{s-3} \Rightarrow Y = \frac{2}{(s-3)(s^2 - s + 2)}. \\
 Y &= \frac{1}{4} \frac{1}{s-3} - \frac{1}{4} \frac{s+2}{s^2 - s + 2} = \frac{1}{4} \frac{1}{s-3} - \frac{1}{4} \frac{s - \frac{1}{2}}{(s - \frac{1}{2})^2 + \frac{7}{4}} - \frac{5}{4} \frac{1}{(s - \frac{1}{2})^2 + \frac{7}{4}}. \\
 \Rightarrow y(t) &= \frac{1}{4} e^{3t} - \frac{1}{4} e^{t/2} \cos \frac{\sqrt{7}t}{2} - \frac{5}{4\sqrt{7}} e^{t/2} \sin \frac{\sqrt{7}t}{2}.
 \end{aligned}$$

(2)

$$\begin{aligned}
 y'' + 4y' + 4y &= t^3 * e^{-2t}, \quad y(0) = 1, \quad y'(0) = 2. \\
 (s^2Y - sy(0) - y'(0)) + 4(sY - y(0)) + 4Y &= \frac{6}{s^4} \cdot \frac{1}{s+2} \\
 (s+2)^2Y &= (s+6) + \frac{6}{s^4(s+2)} \Rightarrow Y = \frac{s+6}{(s+2)^2} + \frac{6}{s^4(s+2)^3}. \\
 \frac{s+6}{(s+2)^2} &= \frac{1}{s+2} + \frac{4}{(s+2)^2}, \\
 \frac{6}{s^4(s+2)^3} &= \frac{-\frac{15}{16}}{s} + \frac{\frac{9}{8}}{s^2} - \frac{\frac{9}{8}}{s^3} + \frac{\frac{3}{4}}{s^4} + \frac{\frac{15}{16}}{s+2} + \frac{\frac{3}{4}}{(s+2)^2} + \frac{\frac{3}{8}}{(s+2)^3}. \\
 \Rightarrow y(t) &= \frac{31}{16} e^{-2t} + \frac{19}{4} t e^{-2t} + \frac{3}{16} t^2 e^{-2t} - \frac{15}{16} + \frac{9}{8} t - \frac{9}{16} t^2 + \frac{1}{8} t^3
 \end{aligned}$$

(3)

$$\begin{aligned}
 y''' + 3y'' + 3y' + y &= 6e^{-t}, \quad y(0) = y'(0) = y''(0) = 0. \\
 (s^3Y - s^2y(0) - sy'(0) - y''(0)) + 3(s^2Y - sy(0) - y'(0)) + 3(sY - y(0)) + Y &= \frac{6}{s+1} \\
 (s^3 + 3s^2 + 3s + 1)Y &= \frac{6}{s+1} \Leftrightarrow (s+1)^3Y = \frac{6}{s+1} \Rightarrow Y = \frac{6}{(s+1)^4}. \\
 \Rightarrow y(t) &= \mathcal{L}^{-1} \left\{ \frac{6}{(s+1)^4} \right\} = t^3 e^{-t}.
 \end{aligned}$$

(4)

$$\begin{aligned}
 y^{(4)} + 2y'' + y &= \sin t, \quad y(0) = y'(0) = y''(0) = y^{(3)}(0) = 0. \\
 (s^4Y + 2s^2Y + Y) &= \frac{1}{s^2+1} \Rightarrow (s^2+1)^2Y = \frac{1}{s^2+1} \Rightarrow Y = \frac{1}{(s^2+1)^3}. \\
 \text{Write } \frac{1}{(s^2+1)^3} &= A \frac{1}{s^2+1} + B \left(-\frac{d}{ds} \frac{s}{s^2+1} \right) + C \left(\frac{d^2}{ds^2} \frac{1}{s^2+1} \right), \\
 A &= \frac{3}{8}, \quad B = -\frac{3}{8}, \quad C = -\frac{1}{8} \\
 \mathcal{L}^{-1} \left\{ \frac{1}{(s^2+1)^3} \right\} &= A \sin t + B t \cos t + C t^2 \sin t \\
 \Rightarrow y(t) &= \frac{1}{8} (3 \sin t - 3t \cos t - t^2 \sin t).
 \end{aligned}$$